

## WARRA, Australia

WMO: 959620

Lat: 43.061S

Lon: 146.704E

Elev: 495

StdP: 95.52

Time Zone: 10.00 (AUS)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | -0.7       | 0.2 | -2.2                          | 3.3 | 2.1  | -1.3 | 3.6 | 2.2  | 12.2                  | 8.2  | 10.9 | 7.6  | 2.0                   | 260  | 0.622 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 10.4                   | 26.3            | 16.3 | 23.9 | 15.7 | 21.6 | 15.0 | 17.6                | 23.7 | 16.6 | 21.9 | 15.7 | 20.3 | 4.1                  | 310  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 15.6                            | 11.8 | 18.4 | 14.6 | 11.0 | 18.1 | 13.7 | 10.4 | 17.4 | 51.4          | 23.6 | 48.5 | 21.8 | 45.5 | 20.4 | 21.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 10.5          | 9.2  | 8.1 | DB                         | -2.1 | 32.0               | 1.0 | 2.6                                                | -2.8 | 33.9       | -3.4 | 35.5       | -3.9 | 37.0       | -4.6 | 38.9 | (4) |
| (5)               |      |     | WB                         | -2.4 | 19.7               | 0.9 | 0.7                                                | -3.0 | 20.2       | -3.5 | 20.7       | -4.0 | 21.1       | -4.6 | 21.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.2    | 13.4  | 13.1  | 11.9  | 9.5   | 7.6   | 6.0   | 5.4   | 5.6   | 7.0   | 8.6   | 10.3  | 11.7 | (6)  |  |
|                                                                        | DBStd       | 4.09   | 3.68  | 3.45  | 3.52  | 2.91  | 2.50  | 2.15  | 1.92  | 2.42  | 2.61  | 3.20  | 3.59  | 3.40 | (7)  |  |
|                                                                        | HDD10.0     | 779    | 8     | 8     | 18    | 42    | 83    | 122   | 142   | 137   | 96    | 66    | 38    | 19   | (8)  |  |
|                                                                        | HDD18.3     | 3368   | 161   | 151   | 202   | 264   | 334   | 400   | 394   | 341   | 302   | 242   | 207   |      | (9)  |  |
|                                                                        | CDD10.0     | 472    | 113   | 94    | 78    | 29    | 8     | 1     | 0     | 2     | 6     | 23    | 46    | 72   | (10) |  |
|                                                                        | CDD18.3     | 17     | 7     | 4     | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 2    | (11) |  |
|                                                                        | CDH23.3     | 274    | 111   | 48    | 42    | 1     | 0     | 0     | 0     | 0     | 0     | 6     | 24    | 42   | (12) |  |
|                                                                        | CDH26.7     | 68     | 33    | 11    | 11    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 11   | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.7    | 3.7   | 3.3   | 3.2   | 3.4   | 3.8   | 3.4   | 3.7   | 3.8   | 3.9   | 3.9   | 3.9   | 3.8  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1421   | 87    | 72    | 85    | 100   | 135   | 134   | 152   | 184   | 153   | 144   | 109   | 109  | (15) |  |
|                                                                        | PrecMax     | 1721   | 165   | 178   | 182   | 213   | 337   | 284   | 295   | 323   | 296   | 315   | 197   | 213  | (16) |  |
|                                                                        | PrecMin     | 1198   | 16    | 30    | 37    | 38    | 49    | 44    | 58    | 61    | 60    | 50    | 24    | 38   | (17) |  |
|                                                                        | PrecStd     | 147    | 43    | 39    | 34    | 39    | 75    | 53    | 51    | 65    | 64    | 57    | 35    | 40   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 30.6  | 28.3  | 27.7  | 21.1  | 16.2  | 12.9  | 11.8  | 14.7  | 17.9  | 23.9  | 26.5  | 28.3 | (19) |  |
|                                                                        |             | MCWB   | 18.0  | 17.2  | 16.6  | 14.1  | 11.8  | 10.7  | 8.9   | 9.6   | 11.7  | 12.9  | 15.6  | 16.3 | (20) |  |
|                                                                        | 2%          | DB     | 26.2  | 24.4  | 23.4  | 18.7  | 14.0  | 11.4  | 10.4  | 12.5  | 14.9  | 19.8  | 23.0  | 23.6 | (21) |  |
|                                                                        |             | MCWB   | 16.4  | 16.5  | 15.7  | 13.3  | 10.8  | 9.4   | 7.9   | 8.7   | 9.8   | 12.3  | 14.9  | 15.4 | (22) |  |
|                                                                        | 5%          | DB     | 23.3  | 21.8  | 20.2  | 16.3  | 12.6  | 10.2  | 9.4   | 10.9  | 13.1  | 17.0  | 19.6  | 20.7 | (23) |  |
|                                                                        |             | MCWB   | 15.7  | 15.6  | 14.9  | 12.2  | 10.0  | 8.3   | 7.4   | 7.9   | 9.2   | 11.2  | 13.6  | 14.6 | (24) |  |
|                                                                        | 10%         | DB     | 20.4  | 19.5  | 17.8  | 14.2  | 11.5  | 9.4   | 8.5   | 9.5   | 11.4  | 14.6  | 16.6  | 18.2 | (25) |  |
|                                                                        |             | MCWB   | 14.6  | 14.7  | 13.9  | 11.3  | 9.4   | 7.8   | 6.9   | 7.3   | 8.4   | 10.5  | 12.3  | 13.3 | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 19.1  | 18.2  | 18.0  | 15.2  | 12.7  | 11.9  | 9.6   | 10.2  | 12.2  | 15.1  | 17.0  | 17.9 | (27) |  |
|                                                                        |             | MCDB   | 27.2  | 25.0  | 23.9  | 19.6  | 14.9  | 12.1  | 10.9  | 12.9  | 16.4  | 18.9  | 22.2  | 23.9 | (28) |  |
|                                                                        | 2%          | WB     | 17.6  | 17.3  | 16.7  | 13.8  | 11.3  | 9.7   | 8.4   | 9.3   | 10.7  | 13.3  | 15.7  | 16.3 | (29) |  |
|                                                                        |             | MCDB   | 24.2  | 22.9  | 21.6  | 17.6  | 13.1  | 10.9  | 9.6   | 11.9  | 13.8  | 17.8  | 21.4  | 21.9 | (30) |  |
|                                                                        | 5%          | WB     | 16.3  | 16.2  | 15.5  | 12.7  | 10.4  | 8.7   | 7.7   | 8.3   | 9.6   | 11.9  | 14.2  | 15.0 | (31) |  |
|                                                                        |             | MCDB   | 21.7  | 20.7  | 19.5  | 15.6  | 12.2  | 9.8   | 9.0   | 10.5  | 12.2  | 16.1  | 18.8  | 19.7 | (32) |  |
|                                                                        | 10%         | WB     | 14.9  | 15.0  | 14.3  | 11.6  | 9.6   | 8.0   | 7.2   | 7.5   | 8.6   | 10.5  | 12.7  | 13.8 | (33) |  |
|                                                                        |             | MCDB   | 19.8  | 18.8  | 17.9  | 13.9  | 11.3  | 9.2   | 8.4   | 9.2   | 11.0  | 14.2  | 16.3  | 17.8 | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.4  | 9.8   | 8.5   | 6.9   | 5.5   | 5.1   | 5.0   | 5.8   | 7.0   | 8.7   | 9.0   | 9.6  | (35) |  |
|                                                                        |             | MCDBR  | 16.6  | 14.3  | 13.1  | 10.1  | 6.5   | 5.8   | 5.8   | 7.7   | 9.9   | 13.6  | 14.8  | 15.2 | (36) |  |
|                                                                        |             | MCWBR  | 8.0   | 6.9   | 6.9   | 6.0   | 4.7   | 4.5   | 4.4   | 5.0   | 6.1   | 7.6   | 7.6   | 7.7  | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 14.5  | 12.7  | 11.6  | 9.3   | 6.2   | 5.2   | 5.4   | 7.1   | 8.9   | 12.0  | 13.1  | 13.3 | (38) |  |
|                                                                        |             | MCWBR  | 7.6   | 6.8   | 6.6   | 6.0   | 5.0   | 4.3   | 4.5   | 4.9   | 6.0   | 7.2   | 7.2   | 7.6  | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.331  | 0.319 | 0.319 | 0.304 | 0.295 | 0.284 | 0.286 | 0.300 | 0.324 | 0.327 | 0.323 | 0.327 |      | (40) |  |
|                                                                        | taud        | 2.529  | 2.581 | 2.571 | 2.598 | 2.602 | 2.616 | 2.615 | 2.581 | 2.493 | 2.496 | 2.531 | 2.530 |      | (41) |  |
|                                                                        | Ebn at Noon | 994    | 980   | 933   | 878   | 813   | 790   | 817   | 869   | 911   | 960   | 997   | 1004  |      | (42) |  |
|                                                                        | Edh at Noon | 108    | 97    | 89    | 73    | 61    | 55    | 59    | 73    | 94    | 104   | 107   | 109   |      | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 6.27   | 5.39  | 3.93  | 2.64  | 1.59  | 1.26  | 1.45  | 2.17  | 3.27  | 4.69  | 5.72  | 6.24  |      | (44) |  |
|                                                                        | RadStd      | 0.46   | 0.39  | 0.24  | 0.15  | 0.12  | 0.11  | 0.09  | 0.16  | 0.17  | 0.27  | 0.44  | 0.39  |      | (45) |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (27 neighbors) | +0.32 | N/A     | N/A    | +0.68   | N/A   | N/A   | N/A         | -100    | +106    | +14     |

Nomenclature: See separate page